

KAMIREDDY NITHIN KUMAR REDDY

AI ENGINEER | SOFTWARE ENGINEER | FULL STACK DEVELOPER

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PROFESSIONAL SUMMARY

Computer Science Engineering graduate with hands-on experience across Generative AI, machine learning, RAG systems, full-stack development, REST APIs and software engineering. Skilled in Python, Java, Spring Boot, React, JavaScript, SQL/MySQL, Git/GitHub, Linux and AWS. Experienced in building AI-powered applications, retrieval-augmented generation workflows, code-analysis systems, predictive maintenance solutions and API-integrated web applications. Strong foundation in OOP, data structures, databases, API integration, problem solving and research.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, HTML5, CSS3
AI & Machine Learning: Generative AI, Machine Learning, NLP, RAG, LLMs, AI Agents, Prompt Engineering, Embeddings, Semantic Retrieval
Backend: Spring Boot, REST APIs, RESTful Services, API Development
Frontend: React.js, JavaScript, HTML5, CSS3, Responsive UI Development
Database: MySQL, SQL, Database Design, Query Development
Cloud & Systems: AWS, Linux, Red Hat Linux, Cloud Computing, Networking Fundamentals
Tools: Git, GitHub, IntelliJ IDEA, Visual Studio Code, Postman, Figma, MySQL, Linux Terminal, AWS, Replit
Core Concepts: Object-Oriented Programming, Data Structures & Algorithms, DBMS, REST Architecture, API Integration, Full-Stack Development

SELECTED PROJECTS

CodeScope AI – AI-Powered Code Analysis Platform Generative AI | Python | LLM

- Developed an AI-powered platform for analyzing software projects and generating structured insights about codebases, technologies, project domains and application architecture.
- Designed workflows to accept project repositories or source directories and generate developer-oriented insights for faster codebase understanding and productivity.
- Applied Generative AI and LLM concepts to software analysis, automated understanding and developer productivity.

RAGForge AI – Retrieval-Augmented Generation Application RAG | Python | LLM

- Built a Retrieval-Augmented Generation application combining document retrieval, contextual information and language-model generation for knowledge-based question answering.
- Applied embeddings, semantic retrieval, prompt engineering and LLM-based generation to improve contextual response relevance.

Predictive Maintenance & Process Intelligence Python | Scikit-learn | Pandas | Chart.js

- Developed an end-to-end machine learning pipeline for analyzing machine failure patterns including tool wear, heat dissipation and overstrain.
- Implemented data preprocessing, correlation analysis, feature-importance analysis, failure-trend analysis and machine learning workflows using Python, Pandas and Scikit-learn.
- Created an interactive HTML5 dashboard with visual analytics for machine failures, correlations, feature importance and process intelligence.

AI Chatbot Web Application Python | AI | REST API | React

- Developed and deployed a responsive AI chatbot supporting real-time user interaction through API-based response generation.
- Integrated frontend and backend/API services to support conversational workflows, dynamic responses and reliable user interaction.

EDUCATION

Saveetha School of Engineering, Chennai	2026
B.E. / B.Tech. – Computer Science Engineering	CGPA: 8.5/10
Narayana Junior College	2022
Intermediate – MPC	89.3%

CERTIFICATIONS

AI & Data: TATA Generative AI Powered Data Analytics; Infosys Artificial Intelligence; IBM Python 101 for Data Science; Infosys Springboard Introduction to Data Science; Infosys Springboard Introduction to Deep Learning; Infosys Springboard Introduction to Natural Language Processing.
Software & Cloud: Oracle Database SQL Certified Specialist; HackerRank SQL (Intermediate); HackerRank Python (Basic); Java Certification – Wipro Training Academy; Getting Started with DevOps on AWS; AWS Planning a Generative AI Project; AWS Introduction to Generative AI: Art of the Possible.

ACHIEVEMENTS

- Published 4 research papers focused on classification accuracy and machine learning model performance.
- Received Best Performance Award and presented technical project work involving AI, IoT, robotics and intelligent systems.
- Selected for Infosys Springboard Virtual Internship 7.0 – Batch 1.